

CFTC Positioning & WTI Crude Oil Spreads

Does managed-money positioning predict M1–M2 spread behaviour — at the regime level and around contract expiry?

DATA PERIOD	CFTC WEEKS	EXPIRY EVENTS	PRODUCT	REPORT DATE
2016 – 2026	545	127	WTI CL (NYMEX)	July 2026

EXECUTIVE SUMMARY

Net managed-money positioning (the difference between long and short contracts held by speculative funds) has a measurable and statistically significant relationship with the WTI M1–M2 spread — the price gap between the front-month and second-month crude oil futures contract. The effect is strongest in high-volatility market regimes, where heavier net-long positioning amplifies both roll compression and post-expiry reversion.

Gross long and short positions analysed separately do not add explanatory power over the net position in predicting roll behaviour. The *relative balance* (long-to-short ratio) is similarly uninformative. This confirms that the *direction* of speculative bias — not the sheer size of open interest on either side — is what moves the spread.

SECTION 1

Data Sources & Key Definitions

CFTC Commitments of Traders (COT) Report

Every week the U.S. Commodity Futures Trading Commission publishes the COT report, which discloses how different categories of market participants are positioned across futures markets. For this analysis the relevant category is **Managed Money** — hedge funds and other large speculative accounts. For each week we extract:

- **MM Gross Longs** — total long contracts held by managed money
- **MM Gross Shorts** — total short contracts held by managed money
- **MM Net Long** — gross longs minus gross shorts (positive = net bullish)

The M1–M2 Spread

WTI crude oil trades on the NYMEX in monthly delivery contracts. The **M1–M2 spread** is the price of the front-month contract (nearest delivery) minus the price of the second-month contract. A positive spread means the market is in *backwardation* (near-term crude more expensive — typically signals supply tightness). A negative spread means *contango* (near-term cheaper — typically signals abundant supply).

Markov Regime States

A Hidden Markov Model classifies each week as either a **High-Volatility (HV)** or **Low-Volatility (LV)** regime based on the historical behaviour of the M1–M2 spread. This classification is used throughout the analysis to test whether CFTC effects differ depending on the type of market environment.

Positioning Z-Score (pos_z_full)

Because the raw net-long count has grown over time as the market expanded (from roughly 200,000 contracts net in 2016 to over 400,000 in 2023), raw numbers are not directly comparable. We normalise by computing a **z-score**: how many standard deviations the current net position is from its long-run average. A z-score of +2 means positioning is unusually bullish by historical standards.

SECTION 2

Phase 1 — Does Positioning Predict the Spread Level?

Tests whether the z-scored net managed-money position predicts the M1–M2 spread across the full 2016–2026 period, and whether the relationship strengthens in high-volatility regimes.

What Was Tested

Three **OLS (Ordinary Least Squares) regressions** were run — one on the full pooled sample, one restricted to High-Volatility weeks, and one restricted to Low-Volatility weeks. OLS is the standard method for asking: "on average, if positioning rises by one unit, how much does the spread change?" All standard errors are adjusted using the **Newey-West method** (lag=2), which corrects for the fact that weekly data points are correlated with nearby weeks — a necessary correction for time-series regressions.

A **flow vs. level test** was also run: does the *week-on-week change* in positioning (flow) predict spreads better than the current positioning *level*? The level (z-score) was found to dominate.

MODEL SPECIFICATION

$M1M2\text{ spread} \sim \beta \times \text{pos_z_full} + \epsilon$

Run separately for: (1) all weeks pooled, (2) HV weeks only, (3) LV weeks only.

Standard errors: Newey-West HAC, lag = 2 weeks.

Table 1 — Phase 1 regression results: M1M2 spread on net positioning z-score

MODEL	B (POS_Z_FULL)	STD. ERROR	P-VALUE	SIGNIFICANCE	INTERPRETATION
Pooled (all weeks)	+0.18	0.083	0.030	**	Significant — more bullish positioning → higher spread
Low-Volatility regime	+0.11	0.089	0.190	—	Not significant in calm markets
High-Volatility regime	+0.31	0.099	0.002	***	Strongly significant — momentum dynamic in volatile periods

Significance codes: *** p<0.01 · ** p<0.05 · * p<0.10 · — not significant

Finding: Net CFTC positioning is a significant predictor of the M1–M2 spread, but almost entirely through the High-Volatility channel. In calm markets, positioning has no reliable directional power. In volatile markets, heavier speculative net-long positioning is associated with a wider backwardation spread — consistent with a momentum/crowding dynamic where funds pile into a trade already in motion.

SECTION 3

Phase 2 — CFTC Positioning Around Contract Expiry

Analyses 127 front-month contract expiry events (CLG16 to CLQ26) to test whether positioning predicts how much the spread compresses during the roll week and how quickly it recovers afterwards.

Background: The Roll Window

Each month, the front-month futures contract approaches its last trading day (the NYMEX Last Trading Day, or LTD — the third business day prior to the 25th calendar day of the preceding month). In the week before expiry, traders who hold the front-month contract must "roll" their position into the next month's contract. This selling pressure on the front month typically *compresses* the M1–M2 spread. After expiry, the spread often *reverts* as the artificial selling pressure lifts.

Key Variables Constructed

- **roll_compression:** M1M2 at end of roll week minus M1M2 at entry (5 days before expiry). Negative means the spread narrowed.

- **roll_reversion**: M1M2 five days after expiry minus M1M2 on expiry day. Positive means the spread recovered.
- **pos_z_full**: CFTC net-long z-score from the last available COT release before the roll window begins.

Stage 1 — Does Positioning Predict Roll Compression?

MODEL SPECIFICATION

roll_compression ~ $\beta \times \text{pos_z_full} + \epsilon$
Run separately for HV and LV regimes. Newey-West SE, lag = 2.

Table 2 — Stage 1: Roll compression on net positioning z-score

MODEL	N	B (POS_Z_FULL)	P-VALUE	SIGNIFICANCE	INTERPRETATION
Pooled	65	-0.57	0.163	—	Not significant
High-Volatility regime	~40	-0.18	0.090	*	Marginally significant — bullish positioning associated with slightly more compression
Low-Volatility regime	~25	flat	—	—	No relationship in calm regimes

Finding: CFTC positioning has only a weak, marginally significant relationship with how much the spread compresses during the roll week in High-Volatility regimes. The roll compression itself appears to be largely mechanical — driven by the expiry calendar, not by where funds are positioned.

Stage 2 — Does Positioning Predict Post-Expiry Reversion?

MODEL SPECIFICATION

roll_reversion ~ $\beta_1 \times \text{pos_z_full} + \beta_2 \times \text{roll_compression} + \epsilon$
roll_compression is included as a control — because the amount of reversion depends partly on how much was compressed in the first place. Newey-West SE, lag = 2. Run separately for HV and LV.

Table 3 — Stage 2: Post-expiry reversion on positioning + roll compression

MODEL	B (POS_Z_FULL)	P (POS_Z)	B (ROLL_COMP)	P (COMP)	R ²
Pooled	+0.65	0.158	-0.77	0.000	0.608
High-Volatility regime	+0.299	0.027	-0.85	0.000	0.621
Low-Volatility regime	flat	—	-0.55	0.000	0.582

Finding: The dominant driver of post-expiry reversion is the compression itself ($\beta \approx -0.85$): on average, roughly 83–85% of whatever the spread compressed during the roll week reverts within five days after expiry. This is consistent and statistically significant across both regimes. In **High-Volatility regimes**, CFTC positioning adds an additional tilt: heavier net-long positioning is associated with stronger reversion ($\beta = +0.299$, $p = 0.027$) — mirroring the Phase 1 momentum finding.

Interaction Test — Is the HV Effect Statistically Different from LV?

MODEL SPECIFICATION

roll_compression ~ **pos_z_full** × **HV_dummy** + ϵ

A single model with an interaction term tests whether the coefficient on **pos_z_full** is significantly different between regimes, rather than simply running two separate regressions and eyeballing the difference.

Interaction result: The interaction term is statistically significant ($p = 0.04$), confirming that the CFTC positioning effect is genuinely different in High-Volatility vs. Low-Volatility regimes — not just a sampling artefact.

Logit Test — Does Positioning Predict the Direction of Compression?

MODEL SPECIFICATION

A **logistic regression** (logit) is used when the outcome is binary rather than continuous. Here: will the spread compress (narrow) or widen during the roll week? The logit model estimates the probability of compression as a function of CFTC positioning. If significant, positioning tells us not just how much the spread moves, but which way.

Finding: The logit is flat in both regimes — CFTC positioning does not predict the *direction* of roll compression. This is consistent with the Stage 1 result: positioning influences the magnitude of spread moves but cannot reliably forecast whether the spread will narrow or widen into expiry.

SECTION 4

Phase 3 — Gross Long vs. Short Decomposition

Tests whether separating gross long and short positions (rather than using only their difference) reveals additional predictive power over roll behaviour. Working sample: 52 expiry events from 2020–2026, where the raw CFTC position files are available.

Motivation

Two expiry windows can have identical net positioning (e.g. 200,000 contracts net long) but very different gross structures:

- **Crowded structure:** 500,000 longs + 300,000 shorts = 200,000 net. Both sides are large; the market is congested.
- **Clean structure:** 220,000 longs + 20,000 shorts = 200,000 net. One-sided with little opposition.

Crowding affects how positions unwind at expiry. If gross is the true driver, the individual legs should carry explanatory power that the net misses.

Variables Constructed

- **mm_long_z:** z-score of gross MM long contracts over the 2020–2026 sample
- **mm_short_z:** z-score of gross MM short contracts
- **ls_ratio_z:** z-score of the long-to-short ratio ($\text{mm_long} \div \text{mm_short}$)

Comparison: Which Specification Explains More Variance?

MODEL SPECIFICATIONS COMPARED

Net: outcome ~ pos_z_full (+ roll_compression for Stage 2)

Gross: outcome ~ mm_long_z + mm_short_z (+ roll_compression for Stage 2)

Ratio: outcome ~ ls_ratio_z (+ roll_compression for Stage 2)

R² measures what fraction of the variation in the outcome each specification explains. Adjusted R² penalises models with more variables so they can be compared fairly.

Table 4 — R² comparison across IV specifications (N = 52)

OUTCOME	IV SPECIFICATION	R ²	ADJ. R ²	VERDICT
STAGE 1: ROLL_COMPRESSION				
Roll compression	Net — pos_z_full	0.102	0.084	Best in class
Roll compression	Gross — mm_long_z + mm_short_z	0.011	−0.029	No explanatory power
Roll compression	L/S Ratio — ls_ratio_z	0.000	−0.020	No explanatory power
STAGE 2: ROLL_REVERSION				
Roll reversion	Net — pos_z_full	0.608	0.592	Marginally best
Roll reversion	Gross — mm_long_z + mm_short_z	0.569	0.542	Comparable to net
Roll reversion	L/S Ratio — ls_ratio_z	0.561	0.543	Comparable to net

Gross Decomposition Coefficients — Stage 1

Table 5 — Stage 1 coefficient table: gross longs and shorts on roll compression

MODEL	B (MM_LONG_Z)	P	B (MM_SHORT_Z)	P	READING
Pooled	-0.007	0.974	-0.092	0.483	Neither leg matters
High-Volatility	+0.058	0.885	-0.276	0.268	Neither significant; directions inconsistent
Low-Volatility	+0.184	0.436	+0.147	0.106	Both positive — directionally inconsistent with HV

Finding: Neither gross longs nor gross shorts can predict the direction or magnitude of roll compression. The coefficients are not only insignificant — they flip signs across regimes, confirming the result is noise, not signal. The net position (longs minus shorts) remains the only CFTC-derived variable with any compression signal.

Gross Decomposition Coefficients — Stage 2

Table 6 — Stage 2 coefficient table: gross longs and shorts on post-expiry reversion

MODEL	B (MM_LONG_Z)	P	B (MM_SHORT_Z)	P	B (ROLL_COMP)	P
Pooled	+0.188	0.069	+0.141	0.063	-0.852	0.000
High-Volatility	+0.261	0.109	+0.223	0.148	-0.855	0.000
Low-Volatility	-0.453	0.021	-0.132	0.039	-0.556	0.000

Finding — Low-Volatility regime only: In calm market conditions, higher gross longs *and* higher gross shorts both independently suppress post-expiry reversion (both significant at the 5% level). This is an insight the net measure cannot reveal: in LV regimes, a large and balanced open interest on both sides indicates the roll is well-hedged — commercial participants have likely taken the other side — and the spread therefore does not need to recover sharply after expiry. High-OI but balanced positioning dampens the mechanical snap-back.

SECTION 5

Synthesis of Findings

Table 7 — Summary of all findings across phases

PHASE	QUESTION	VARIABLE TESTED	HV RESULT	LV RESULT
Phase 1	Does positioning predict the spread level?	pos_z_full → M1M2	Significant (**)	Not significant
Phase 2 — S1	Does positioning predict roll compression?	pos_z_full → roll_comp	Marginal (*)	No effect
Phase 2 — S2	Does positioning predict post-expiry reversion?	pos_z_full → roll_rev	Significant (**)	Mechanical only
Phase 2 — Logit	Does positioning predict compression direction?	pos_z_full → P(compress)	No	No
Phase 3 — S1	Do gross longs/shorts predict compression?	mm_long_z + mm_short_z → roll_comp	No	No
Phase 3 — S2	Do gross longs/shorts predict reversion?	mm_long_z + mm_short_z → roll_rev	Not significant	Both negative (**)
Phase 3	Does the L/S ratio add information?	ls_ratio_z → both outcomes	No	No

Three Conclusions

1. CFTC net positioning matters, but only in volatile markets. The relationship between managed-money sentiment and the M1–M2 spread is real but regime-conditional. In calm periods, positioning is noise. In volatile periods, it is signal — and specifically a momentum signal, where the spread tends to move in the same direction as speculative bias.

2. The roll mechanism is mostly calendar-driven, not positioning-driven. Approximately 83% of whatever the spread compresses during the roll week reverts within five days of expiry — regardless of where funds are positioned. This compression-reversion cycle is structural. CFTC positioning adds a directional tilt to the reversion in High-Volatility regimes but does not create or prevent the underlying mechanical move.

3. Gross decomposition reveals one incremental insight. Knowing longs and shorts separately does not improve compression prediction. But in Low-Volatility regimes, high gross open interest on both sides (large but balanced) suppresses post-expiry reversion — a signal that is invisible in the net figure and consistent with well-hedged rolls that require no spread recovery.

Data: CFTC COT reports 2016–2026 (545 weekly observations); NYMEX WTI front-month and second-month prices (daily, 2021–2026); 15-minute OHLCV bars for M1–M2 spread construction.

Methods: All regressions use OLS. Standard errors are Newey-West HAC corrected (lag = 2) throughout to account for serial correlation in weekly data. Regime classification via Hidden Markov Model (2 states). Expiry dates computed from NYMEX LTD rules with a custom holiday calendar that includes NYMEX-specific early-close days (Black Friday, Christmas Eve).

Coverage note: Gross long/short data available from 2020 only (raw CFTC txt files). Pre-2020 CFTC data sourced from a curated Excel file containing net positioning only. Phase 3 working sample is therefore 52 of 127 expiry events.

Significance: *** $p < 0.01$ · ** $p < 0.05$ · * $p < 0.10$